Buzzard cluster mock data vector with optical-selection effects: forward model, analytical and empirical $\mathcal{B}_{\rm sel}$ closures

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Abstract

This document is the technical reference for `MockDataVector.ipynb`: a Buzzard-halo DES Y3-like cluster mock that layers the Costanzi+2026 (C26) optical selection systematics on top of the raw simulation outputs, on the DES Y1 cluster binning ($\lambda \in \{20, 30, 45, 60, \infty\}$, $z \in \{[0.20, 0.33), [0.37, 0.50), [0.50, 0.65)\}$ — the standard DES Y1 redshift edges with the upper edge of z_0 pulled in from $0.35 \to 0.33$ and the lower edge of z_1 pushed out from $0.35 \to 0.37$ to skip the Buzzard $0.33 \le z < 0.37$ box-junction seam). The data vector is constructed along *two parallel routes* that share the same Costanzi+2019 (C19) forward model upstream and the same lensing geometry: an **analytical** closure based on the C26 Appendix C Eq. C1 fitting function applied to the C19-binned stack (cheap enough for an MCMC chain), and an **empirical** ratio that stacks on the catalog redMaPPer outputs (`LAMBDA_CHISQ`, `Z_LAMBDA`) and divides by a $(\log M, z^{\rm tr})$-matched random reference (Wu+2022, Costanzi+2026). The notebook also produces a binned number-count vector $N(\lambda^{\rm ob}, z^{\rm ob})$. Member-galaxy contamination (boost factor) is injected from the published DES Y1 measurements (McClintock+2019), so the saved final shear $\gamma_t^{\rm mock,obs} = \Delta\Sigma_{\rm obs,C1}/B_{\rm Y1} \cdot \Sigma_{\rm crit}^{-1}$ mimics the raw uncorrected DES Y1 data vector. All outputs are saved to `output/MockDataVector.npz`; the headline plots embedded in this document live in `output/figs/`.

Contents

1 Pipeline topology

The notebook executes the following stages in order:

halo_load → photoz_proxy → sample_lambda_true → sample_lambda_obs → lensing_inputs → number_counts → analytical_C1 → empirical_mass_match → save_npz

The first four stages are the C19 forward model: select host halos above $\log_{10} M_{\rm vir} = 13$ in $z \in [0.20, 0.65]$ (with the Buzzard $0.33 \leq z < 0.37$ box seam excised), add a DES Y3-like Gaussian photo-z scatter to produce $z^{\rm ob}$, and draw $\lambda^{\rm tr}$ and $\lambda^{\rm ob}$ from the C26/C19 kernels. lensing_inputs centralises the per-halo profiles and $\Sigma_{\rm crit}^{-1}$ table, plus the catalog redMaPPer columns (LAMBDA_CHISQ, Z_LAMBDA) used by the empirical route. The two lensing-data-vector stages are independent and produce their outputs on the same DES Y1 binning so the user can compare them side-by-side. The final stage persists everything to output/MockDataVector.npz.

2 Shared conventions

Cosmology. FlatLambdaCDM(H0=70, Om0=0.3, Ob0=0.046, Tcmb0=2.725) via astropy.cosmology, matching the Buzzard simulation cosmology used by the upstream halo catalogue.

Units. Per-halo $\Sigma, \Delta\Sigma$ are stored in physical $M_\odot/{\rm pc}^2$; radii rp_phys_mpc (from src/radial_bins_phys_mpc.py) are in physical Mpc, with $N_R = 15$ log-spaced bins between 0.0323 and 30 Mpc. Halo masses Mvir are in $h^{-1}\,M_\odot$.

Halo selection. Mirrors 0-MakeMock.ipynb: pid == -1 (host halos only), $0 \leq \cos i \leq 1$ (measured), drop the $0.33 \leq z < 0.37$ simulation-box seam, then require $0.20 \leq z^{\rm tr} \leq 0.65$ and $\log_{10} M_{\rm vir} \geq 13$. Yields $\sim 5.83 \times 10^5$ halos.

DES Y1 binning (used everywhere). Both abundance and lensing bins follow the DES Y1 cluster edges (Costanzi+2019, McClintock+2019), with the central two redshift edges shifted inward by ± 0.02 to skip the Buzzard box-junction seam at $0.33 \leq z < 0.37$:

$$\texttt{LBDBINS} = \{20,\,30,\,45,\,60,\,500\}, \qquad \texttt{ZMIN\_LIST} = \{0.20,\,0.37,\,0.50\}, \qquad \texttt{ZMAX\_LIST} = \{0.33,\,0.50,\,0.65\}.$$

The 500 caps the open $[60, \infty)$ richness range. There are $N_\lambda \times N_z = 4 \times 3 = 12$ bins. The seam volume is $\sim 5\%$ of z_0 and $\sim 13\%$ of z_1 on the published Y1 edges; pulling the bin edges inward eliminates the bias on $N(\lambda^{\rm ob}, z^{\rm ob})$ at the cost of a small mismatch with the Y1-published bin ranges (negligible on $B_{\rm sel}$, $B_{\rm data}$, and the shear data vector since these are intensive quantities).

Photo-z proxy. Per-halo Gaussian

$$z^{\rm ob} = z^{\rm tr} + \sigma_z(1 + z^{\rm tr})\,\mathcal{N}(0,1), \quad \sigma_z = 0.01, \tag{1}$$

clipped at $z^{\rm ob} \geq 0.02$. Stand-in for the DES Y3 redshift kernel of Myles+2021; captures the dominant scatter but not the tails.

Random seeds. A single `numpy.random.default_rng(seed=42)` is shared across the forward model. The MC-vs-PDF sanity histogram uses `seed=123`; the §8.1 bootstrap uses `seed=0`.

3 The intrinsic mass–richness relation (C26 Eq. 15)

Path `src/costanzi_selection.py` (`l_sat`, `l_tr`, `sample_lambda_true`).

The intrinsic richness $\lambda^{\rm tr}$ of a halo (the true number of red-sequence satellite galaxies hosted by the halo, plus the central galaxy) is modelled as a halo central plus a Poisson satellite count. The mean satellite count $\bar{\lambda}_{\rm sat}(M, z)$ follows C26 Eq. 15,

$$\bar{\lambda}_{\rm sat}(M, z) = \left(\frac{M - M_{\rm min}}{M_{\rm pivot}}\right)^{\alpha} \left(\frac{1+z}{1+z_{\rm piv}}\right)^{\epsilon}, \tag{2}$$

clipped to zero for $M < M_{\rm min}$. The mean intrinsic richness adds the central galaxy when the halo is above threshold:

$$\langle \lambda^{\rm tr} | M, z\rangle = \begin{cases} 1 + \bar{\lambda}_{\rm sat}(M, z), & M \geq M_{\rm min}, \\ 0, & M < M_{\rm min}. \end{cases} \tag{3}$$

The DES Y1 NC+3×2pt best-fit constants, hard-coded in `src/costanzi_selection.py`:

$$\log_{10} M_{\rm min} = 11.385, \quad \alpha = 0.859, \quad \log_{10} M_1 = 12.696, \quad \sigma_{\rm intr} = 0.181, \quad \epsilon = 0.284, \quad z_{\rm piv} = 0.4544,$$

with $M_{\rm pivot} = M_1 - M_{\rm min}$.

The per-halo $\lambda^{\rm tr}$ value is drawn (C26 Sec. III) by *(i)* broadening the Poisson rate $\bar{\lambda}_{\rm sat}$ with a multiplicative log-normal of width $\sigma_{\rm intr}$, *(ii)* drawing an integer satellite count, and *(iii)* adding 1 for the central galaxy if the halo is above $M_{\rm min}$:

$$\tilde{\lambda}_{\rm sat} = \bar{\lambda}_{\rm sat}(M, z)\, \exp\!\left[\sigma_{\rm intr}\,\mathcal{N}(0,1) - \tfrac{1}{2}\sigma^2_{\rm intr}\right], \qquad N_{\rm sat} \sim {\rm Poisson}(\tilde{\lambda}_{\rm sat}), \qquad \lambda^{\rm tr} = N_{\rm cen}(M) + N_{\rm sat},$$

where $N_{\rm cen}(M) = 1$ if $M \geq M_{\rm min}$ and 0 otherwise.

4 The projection kernel (C19 Eq. 6)

Path `src/costanzi_selection.py` (`projection_params`, `sample_lambda_obs`). C19 Eq. 6 is a Gaussian-core + exponentially-modified-Gaussian (EMG) mixture:

$$P(\lambda^{\rm ob}|\lambda^{\rm tr}, z) = (1 - f_{\rm prj})\,\mathcal{N}(\mu, \sigma) + f_{\rm prj}\,{\rm EMG}(\mu, \sigma, \tau). \tag{4}$$

The four parameters $\{\mu, \sigma, \tau, f_{\rm prj}\}$ depend on $\lambda^{\rm tr}$ with z-dependent coefficients calibrated from SDSS redMaPPer (C19 Appendix A, Eq. A8):

$$\begin{aligned} \mu(\lambda^{\rm tr}, z) &= a_\mu(z) + b_\mu(z)\,\lambda^{\rm tr}, & \sigma(\lambda^{\rm tr}, z) &= b_\sigma(z)\,(\lambda^{\rm tr})^{a_\sigma(z)}, \\ \tau(\lambda^{\rm tr}, z) &= b_\tau(z)\,(\lambda^{\rm tr})^{-a_\tau(z)}, & f_{\rm prj}(\lambda^{\rm tr}, z) &= b_{f_{\rm prj}}(z)\,\left(1 + e^{-\lambda^{\rm tr}}\right)^{-a_{f_{\rm prj}}(z)}. \end{aligned} \tag{5}$$

The 15 rows of `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` are interpreted as posterior samples and collapsed to the posterior mean via `load_prj_posterior_mean`.

The mixture is sampled directly using the EMG $= \mathcal{N}(\mu, \sigma) + {\rm Exp}(1/\tau)$ representation, avoiding a closed-form EMG density evaluation in the sampling path. The closed form is used only for the sanity overlay below.

Sanity check 1 — MC draws vs analytical kernel. Histogram repeated `sample_lambda_obs` draws at fixed $(\lambda^{\rm tr}, z)$ and overlay the closed-form C19 PDF; they should agree within Poisson noise.

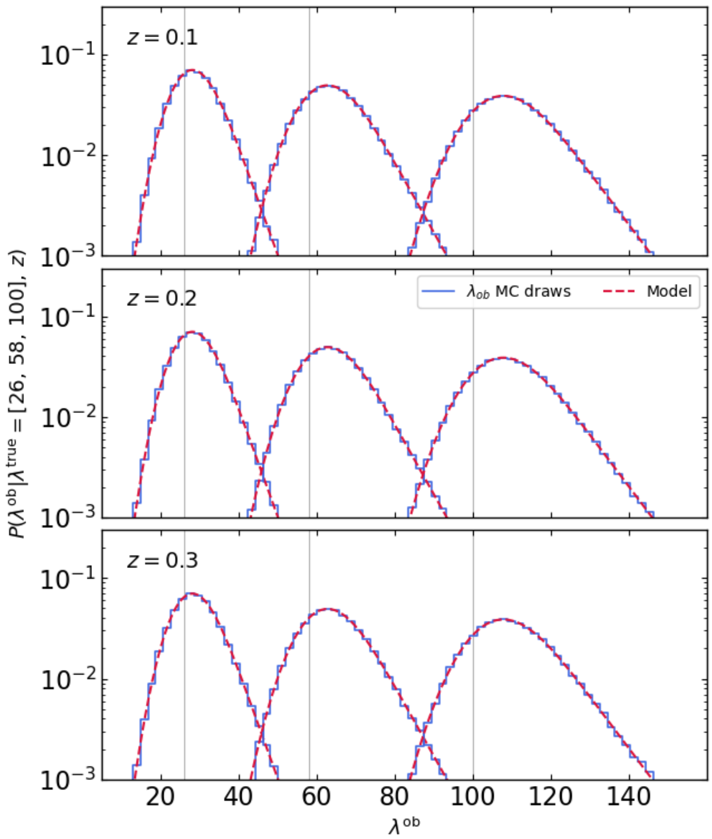

Figure 1: MC draws of $\lambda^{\rm ob}$ from `sample_lambda_obs` (blue histograms) overlaid with the closed-form PDF of Eq. 4 (red dashed) at three reference redshifts and three intrinsic richnesses. The two agree within Poisson noise across the bulk and the tail.

Sanity check 2 — Gaussian core + EMG tail decomposition. Same kernel as Eq. 4, split into the two physical components. At large $\lambda^{\rm tr}$ the EMG tail dominates the high-$\lambda^{\rm ob}$ side and drives the projection bias; this is what the analytical $\mathcal{B}_{\rm sel,C1}$ of §6 fits.

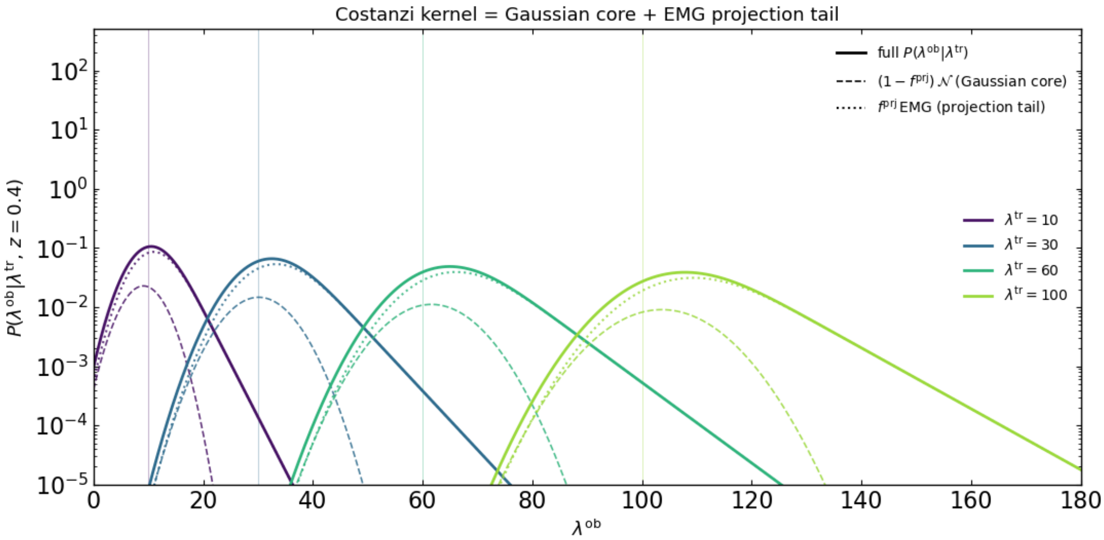

Figure 2: Decomposition of the kernel of Eq. 4 into its $(1-f_{\rm prj})\mathcal{N}$ Gaussian core (dashed) and $f_{\rm prj}$ EMG projection tail (dotted) at $z=0.4$ for four intrinsic richnesses $\lambda^{\rm tr}\in\{10,30,60,100\}$. The EMG tail rises with $\lambda^{\rm tr}$ and drives the optical-selection bias on the lensing data vector.

5 Output 1 — $N(\lambda^{\rm ob}, z^{\rm ob})$ number counts

We obtain $N(\lambda^{\rm ob}, z^{\rm ob})$ by counting the mock halos that fall into each $(\lambda^{\rm ob}, z^{\rm ob})$ bin on the DES Y1 grid. Both axes are forward-model outputs rather than catalog redMaPPer columns: the C19 sampler depends only on (M, z) and carries no per-halo LSS information, which matches the inputs of the analytical $\mathcal{B}_{\rm sel,C1}(\lambda^{\rm ob}, z^{\rm ob})$ of §6 and keeps the abundance and lensing sides of the data vector in the same C26/C19 model world. $N(\lambda^{\rm ob}, z^{\rm ob})$ is therefore the correct entry for a C26 likelihood integrating Eq. 2 into Eq. 4.

6 Lensing data vector — ANALYTICAL route (Matteo / C26 Eq. C1)

Binning quantity. The C19 forward-model $(\lambda^{\rm ob}, z^{\rm ob})$. Each $\lambda^{\rm ob}$ is an independent draw from the kernel of Eq. 4 conditioned only on $(\lambda^{\rm tr}, z)$: the kernel encodes the LSS-driven projection statistics through its EMG tail at the population level, but per-halo $\lambda^{\rm ob}$ values are not correlated with the actual line-of-sight structure around each halo and therefore carry no information correlated with the per-halo lensing signal.

Selection bias. redMaPPer counts red-sequence galaxies inside a projected aperture with no line-of-sight resolution, so projected LSS preferentially boosts $\lambda^{\rm RM}$ for halos in over-dense filaments and lifts the stacked $\Delta\Sigma(R)$ above the mass-only expectation. C26 Appendix C captures this lift with a 5-parameter fitting function (no integration); Matteo (priv. comm.) reports the same form fits the bias on $\Delta\Sigma$:

$$\mathcal{B}^{\Delta\Sigma}_{\rm sel,C1}(R) = A\left(\frac{R}{R_0}\right)^{\alpha}\left[1+\left(\frac{R}{R_0}\right)^{\gamma}\right]^{(\beta-\alpha)/\gamma} + C, \tag{6}$$

with $(A,\alpha,\beta,\gamma,C) = (0.12,\ 4.11,\ -0.18,\ 1.82,\ 1.00)$, $R_0 = R_\lambda(\lambda^{\rm ob})\,(1+z^{\rm ob})$, and $R_\lambda = (\lambda^{\rm ob}/100)^{0.2}\,h^{-1}\,{\rm Mpc}$ (C26 Appendix B).

The C26 paper writes the published form for Σ with a fixed $+1$ rather than a free C ($C = 1$ limit of Eq. 6); we plot both flavours for comparison. The data vector is

$$\Delta\Sigma_{\rm obs}(R) = \mathcal{B}^{\Delta\Sigma}_{\rm sel,C1}(R)\,\Delta\Sigma_{\rm stack}(R), \qquad \langle\gamma_t(R)\rangle = \Delta\Sigma_{\rm obs}(R)\,\Sigma^{-1}_{\rm crit}\Big(\langle z^{\rm ob}\rangle_{\rm bin}\Big)\,, \tag{7}$$

with $\Delta\Sigma_{\rm stack}$ the unweighted mean of `DeltaSigma` across the C19-binned halos in each $(\lambda^{\rm ob}, z^{\rm ob})$ cell.

Why a fitting function? An MCMC chain evaluates the lensing model $\sim 10^4$–10^6 times, so the per-step cost matters. Computing the full Costanzi-2026 prediction from scratch requires nested integrals over halo mass and redshift at every step — prohibitive at chain length. Equation (6) replaces those integrals with five constants per bin, cutting the cost from seconds to microseconds without changing the answer.

Selection-bias ratio $\mathcal{B}^{\Delta\Sigma}_{\rm sel,C1}(R)$ The fitting function evaluated at the bin-mean $(\langle\lambda^{\rm ob}\rangle, \langle z^{\rm ob}\rangle)$ for every $(\lambda^{\rm ob}, z^{\rm ob})$ cell, with the published Σ-flavour overlay.

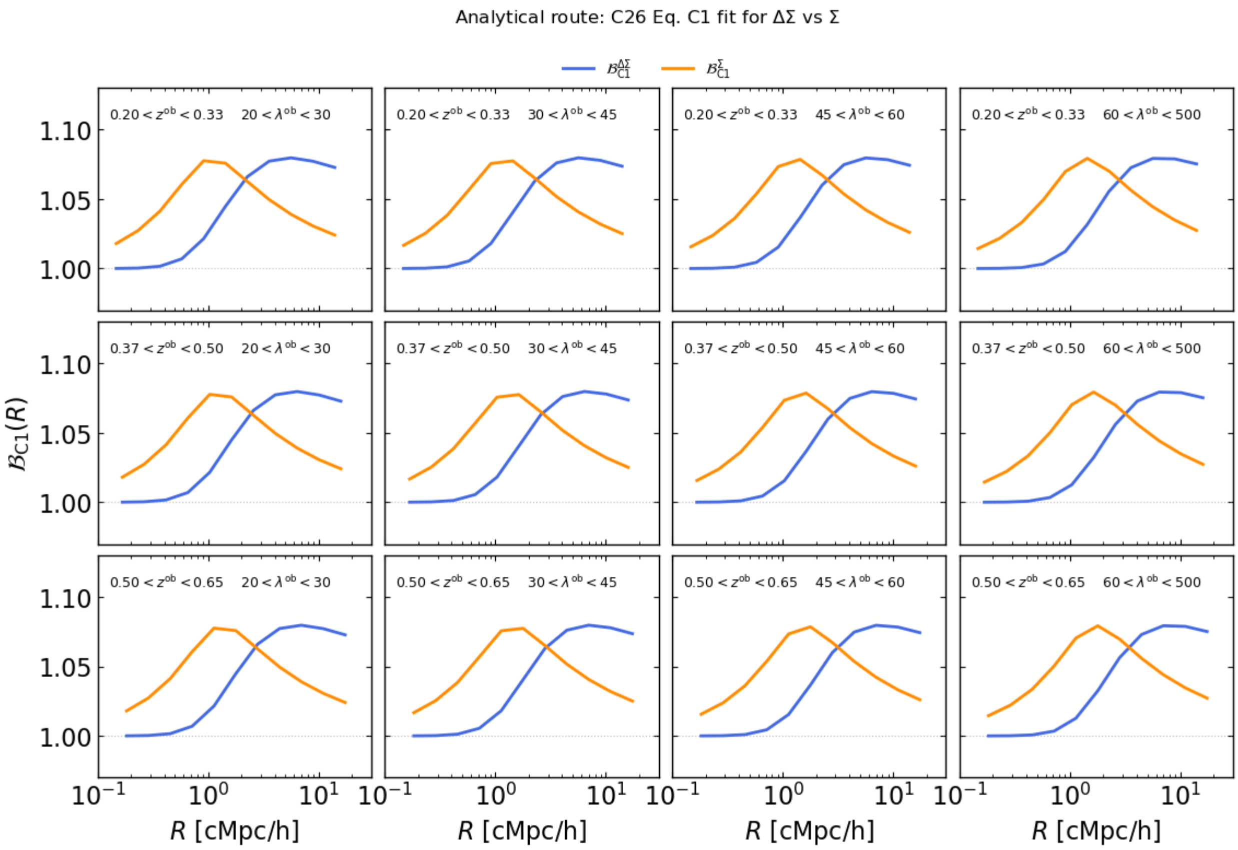

Figure 3: Analytical route: C26 Eq. C1 selection-bias ratio per $(\lambda^{\rm ob}, z^{\rm ob})$ bin on the DES Y1 grid. Blue: $\mathcal{B}^{\Delta\Sigma}_{\rm sel,C1}$ (used in the data vector). Orange: $\mathcal{B}^{\Sigma}_{\rm sel,C1}$ overlay. x-axis is comoving R in h^{-1} Mpc to match the C26 convention; y-axis is the bias factor.

6.1 Boost factor: member-galaxy contamination $B(R)$

A real DES Y1 measurement of $\Delta\Sigma$ is diluted by cluster-member galaxies that survive photo-z cuts and end up in the lensing source catalog. These members carry no shear signal and reduce the measured profile by the boost factor $B(R) \equiv 1/(1-f_{\rm cl}) \geq 1$ (McClintock+2019 §5.3.1).

$$B(R;\, r_s, b_0) = 1 + b_0\,\frac{1-f(x)}{x^2-1}, \qquad x = R/r_s,$$

$$f(x) = \begin{cases} \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}, & x > 1 \\ \dfrac{\operatorname{arctanh}\sqrt{1-x^2}}{\sqrt{1-x^2}}, & x < 1 \end{cases} \tag{8}$$

To make the saved data vector mock the *raw uncorrected* DES Y1 measurement, both lensing routes have their final shear stack divided by $B(R)$:

$$\langle\gamma_t\rangle^{\rm mock,obs} = \frac{\langle\gamma_t\rangle^{\rm obs}}{B_{\rm Y1\,data}(R)} \tag{9}$$

Where $B_{\rm Y1\,data}(R)$ comes from. The DES Y1 cluster analysis released measured boost-factor profiles per richness/redshift bin, located at `data/boost_factor/profiles/` (`full-unblind-v2-mcal-zmix_y1clust_l{l}_z{` The files are tabulated on an 11-point physical-Mpc R grid (0.166 to 15.80 Mpc); we adopt this grid as the global radial axis for the saved data vector. The 7 cosmology-side richness bins `10..16` include $\lambda < 20$ ranges; only `13..16` match our 4 lensing bins $\lambda \in \{[20,30),[30,45),[45,60),[60,\infty)\}$. The parametric McClintock+2019 Table B1 fit is also computed (cell 42) for sanity but is not used in the saved output.

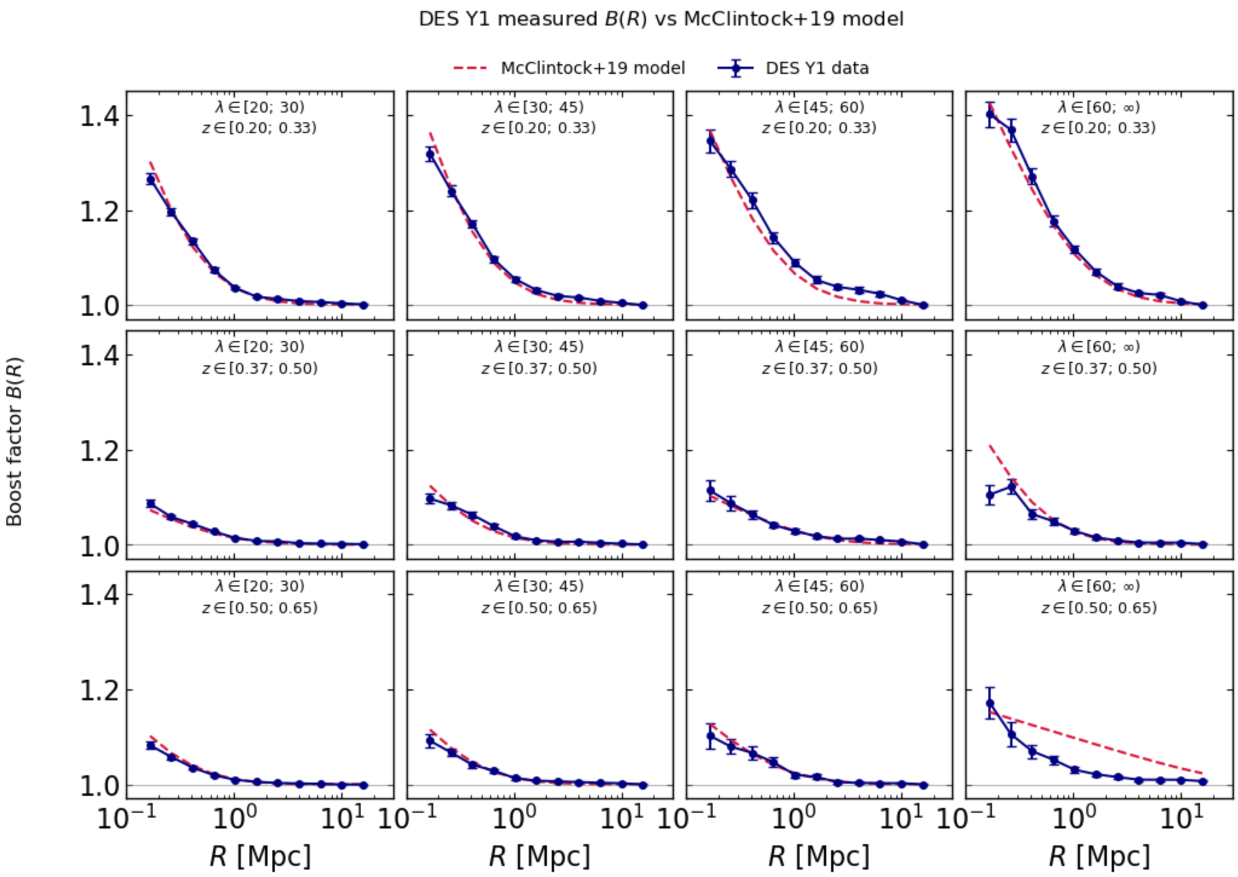

Figure 4: Measured DES Y1 boost factor $B(R)$ per $(\lambda^{\rm RM}, z^{\rm RM})$ bin with reported errors. Small-R amplitude rises with richness and falls with redshift, as expected for a red-sequence-driven member contamination.

7 Lensing data vector — EMPIRICAL route (Heidi / mass-matched ratio)

Binning quantity. The catalog redMaPPer outputs $(\lambda^{\rm RM}, z^{\rm RM})$, i.e. LAMBDA_CHISQ and Z_LAMBDA. These come from running the redMaPPer cluster finder on the Buzzard photometric catalog and *do* carry per-halo LSS information.

Selection bias. Built directly from the catalog as the ratio of the redMaPPer-selected stack to a $(\log M, z^{\rm tr})$-matched random reference, following Wu+2022 method iii (src/stacked_profile_weighted_by_mass_redsh
For each $(\lambda^{\rm RM}, z^{\rm RM})$ bin:

$$\langle \Delta\Sigma(R) \rangle_{\rm rnd} = \frac{\sum_c w_c \, \langle \Delta\Sigma(R) | c \rangle_{\rm full\,mock}}{\sum_c w_c}, \qquad w_c = N_c^{\rm sel}, \tag{10}$$

summed over $(\log_{10} M, z^{\rm tr})$ cells of width $(\Delta \log M, \Delta z) = (0.1, 0.05)$. The cell averages are taken over the *full* mock, so the reference resamples halos that share the selected sample's mass and redshift but were never preferentially picked for their environment. The empirical bias is

$$\mathcal{B}_{\rm sel}^{\rm emp}(R) = \frac{\langle \Delta\Sigma(R) \rangle_{\rm sel}}{\langle \Delta\Sigma(R) \rangle_{\rm rnd}}.$$

No fitting function, no $R \to 0$ extrapolation, both numerator and denominator come straight from DeltaSigma.

Bootstrap. The reported $\mathcal{B}_{\rm sel}^{\rm emp}(R)$ is the **bootstrap mean** over $N_{\rm boot} = 50$ resamples of halos within each bin (with mass-matched reference rebuilt at every step), capturing the statistical uncertainty driven by finite halo count. The shaded bands in the panel grids are the bootstrap standard deviation; the saved file keeps only the bootstrap mean.

7.1 Selection-bias ratio $\mathcal{B}_{\rm sel}^{\rm emp}(R)$ with bootstrap bands

We evaluate $\mathcal{B}_{\rm sel}^{\rm emp}(R)$ on the redMaPPer-binned $(\lambda^{\rm RM}, z^{\rm RM})$ grid using Eq. (10) and report the bootstrap mean and 1σ band per bin. The expected signature is the bell-shaped excess peaking near $R \sim 1$ pMpc that Wu+2022 (Fig. 2) identify as the LSS-environment-driven selection bias: at smaller scales the one-halo term dominates and the ratio approaches unity, while at larger scales the projected over-density that boosted $\lambda^{\rm RM}$ falls outside the lensing aperture and the ratio relaxes back toward one.

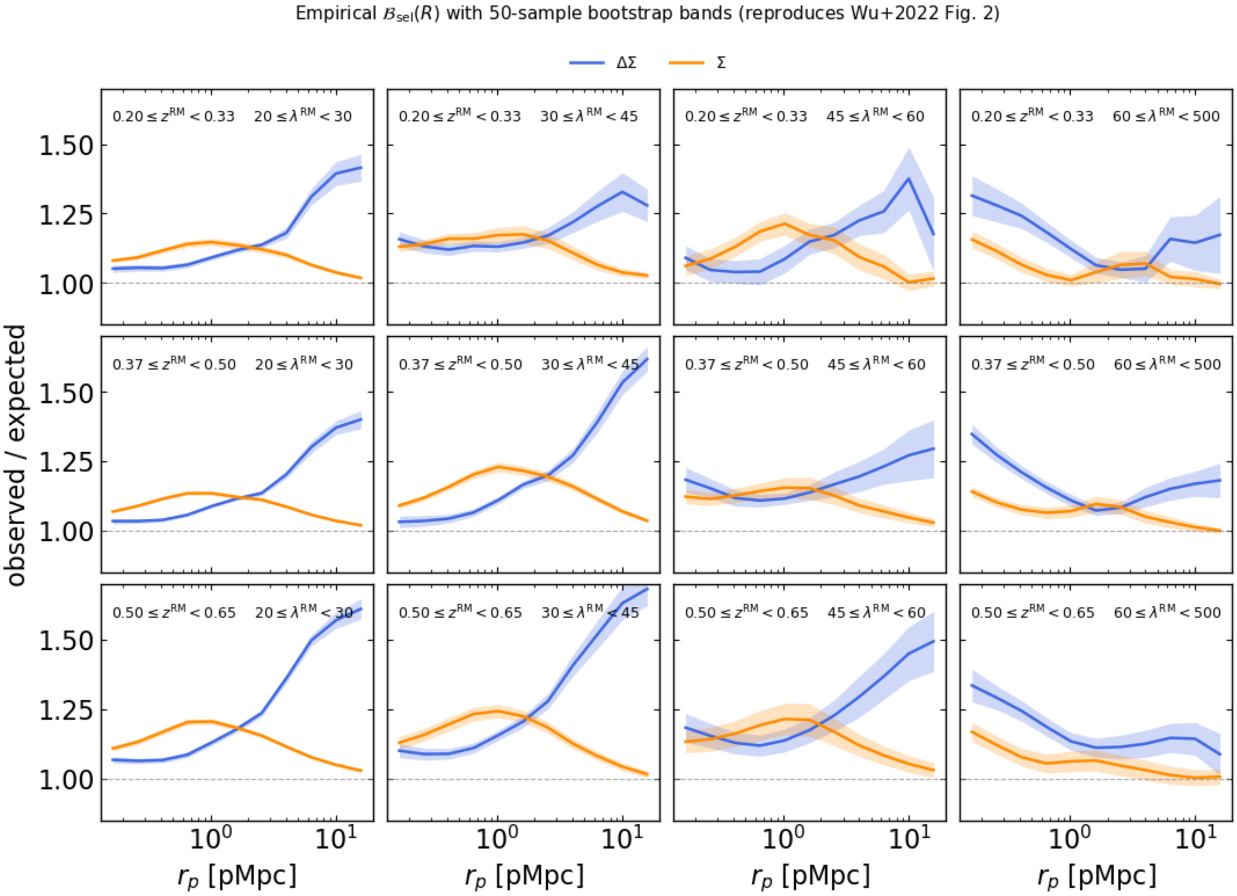

Figure 5: Empirical route: $\mathcal{B}_{\rm sel}^{\rm emp}(R)$ per $(\lambda^{\rm RM}, z^{\rm RM})$ bin on the DES Y1 grid, with $N_{\rm boot} = 50$ bootstrap bands. Blue: $\Delta\Sigma$ ratio. Orange: Σ ratio (overlay). Reproduces Wu+2022 Fig. 2: the bell-shaped peak around $R \sim 1$ pMpc is the LSS-environment-driven selection bias signal.

7.2 Null test: replace $\lambda^{\rm RM}$ with the C19 sampler

If we re-bin on the C19 forward-model $(\lambda^{\rm ob}, z^{\rm ob})$ instead of the catalog $(\lambda^{\rm RM}, z^{\rm RM})$, the same mass-matched-random pipeline should return unity at every R, because the C19 sampler is by construction blind to per-halo LSS. Any residual deviation from 1 is the bias of the matching procedure itself, not a physical signal.

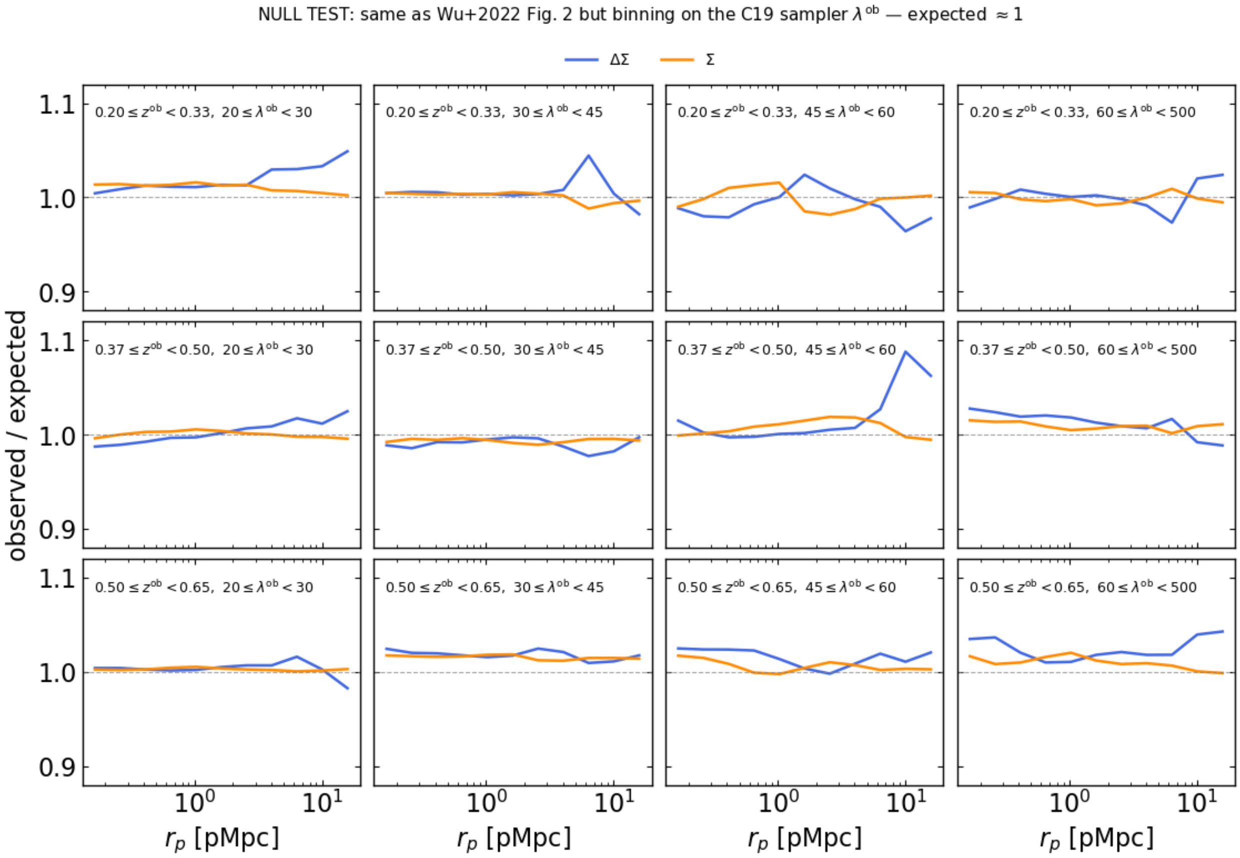

Figure 6: Null test: same Wu+22 panel grid but binning on the C19 sampler ($\lambda^{\rm ob}, z^{\rm ob}$) rather than the catalog ($\lambda^{\rm RM}, z^{\rm RM}$). The bell collapses to ≈ 1 across all bins, confirming the $R \sim 1$ pMpc peak in Fig. 5 is genuinely LSS-driven.

7.3 Tangential shear: redMaPPer-selected vs mass-matched random

The bias ratio of §7.1 is most directly seen by comparing the two tangential-shear stacks that enter its numerator and denominator. The mass-matched random reference is the γ_t a hypothetical unbiased selector would measure on halos of the same $(\log M, z^{\rm tr})$ distribution; the redMaPPer-selected stack is the γ_t an actual cluster finder produces. Their offset is the observable counterpart of $\mathcal{B}^{\rm emp}_{\rm sel}(R)$ and the quantity an MCMC must reproduce.

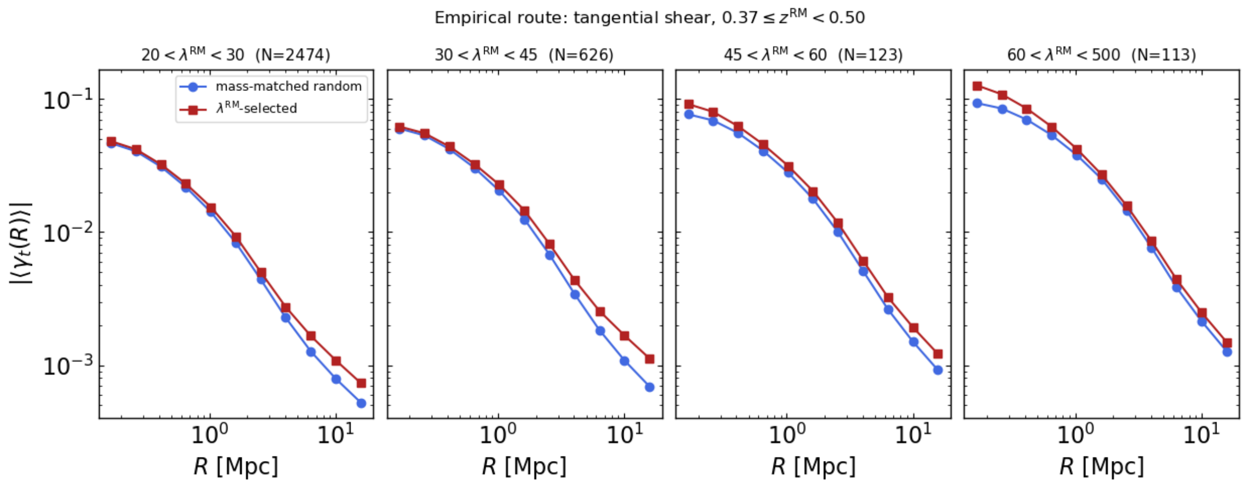

Figure 7: Empirical route: per-bin tangential shear in the middle redshift slice $z^{\rm RM} \in [0.37, 0.50)$. Blue circles: γ_t from the mass-matched random reference. Red squares: γ_t from the $\lambda^{\rm RM}$-selected stack (the observed signal). The selected curve sits above the reference by the $\mathcal{B}_{\rm sel}^{\rm emp}(R)$ factor of Fig. 5.

8 Output archive

The final notebook cell writes both routes plus bin metadata to a single `output/MockDataVector.npz`. With $N_\lambda = 4$, $N_z = 3$, $N_R = 11$ (DES Y1 boost-factor profile grid):

Key	Shape	Content
Bin metadata		
`radii_phys_mpc`	$(N_R,)$	11-pt DES Y1 boost grid (physical Mpc; 0.166–15.80)
`lambda_bins`	$(N_\lambda + 1,)$	DES Y1 richness edges $\{20, 30, 45, 60, 500\}$
`z_bin_min`	$(N_z,)$	lower-redshift edges
`z_bin_max`	$(N_z,)$	upper-redshift edges
Number counts (DES Y1 expectation)		
`NC`	(N_λ, N_z)	$N(\lambda^{\rm ob}, z^{\rm ob})$, Buzzard counts rescaled by $\Omega_{\rm Y1}(z)$
Analytical route, C19-binned		
`gamma_t_stack_C19`	(N_λ, N_z, N_R)	stacked γ_t, no correction
`gamma_t_obs_C1`	(N_λ, N_z, N_R)	stack $\times\, \mathcal{B}_{\rm sel,C1}^{\Delta\Sigma}$
`B_sel_C1`	(N_λ, N_z, N_R)	analytical $\mathcal{B}_{\rm sel,C1}^{\Delta\Sigma}(R)$
Empirical route, RM-binned		
`gamma_t_stack_RM`	(N_λ, N_z, N_R)	$(\log M, z^{\rm tr})$-matched random reference γ_t
`gamma_t_obs_RM`	(N_λ, N_z, N_R)	$\lambda^{\rm RM}$-selected stack γ_t (the observed signal)
`B_sel_emp`	(N_λ, N_z, N_R)	bootstrap-mean $\mathcal{B}_{\rm sel}^{\rm emp}(R)$
DES Y1 boost factor + final mock-observed shears (§6.1)		
`B_data`	(N_λ, N_z, N_R)	measured DES Y1 $B(R)$
`B_data_err`	(N_λ, N_z, N_R)	1-σ error per radial bin
`B_data_cov`	$(N_\lambda, N_z, N_R, N_R)$	full radial covariance matrix per bin
`gamma_t_mock_obs_C1`	(N_λ, N_z, N_R)	analytical (Matteo) shear with Y1 boost dilution
`gamma_t_mock_obs_RM`	(N_λ, N_z, N_R)	empirical (Heidi) shear with Y1 boost dilution

Table 1: Output-archive layout. Default shapes are $(4, 3)$ for the counts and $(4, 3, 15)$ for the radial vectors.

9 Known simplifications

- **Photo-z kernel.** The Gaussian $\sigma_z = 0.01\,(1+z^{\rm tr})$ proxy is a stand-in for the redshift- and colour-dependent kernel of Myles+2021. It captures the dominant scatter but misses the tails that drive the high-z outlier rate.
- **Boost factor uses DES Y1, not Y3, measurements.** §6.1 injects $B(R)$ from McClintock+2019 Y1 profiles, which is the closest released measurement matching our cluster binning. A Y3 boost-factor measurement is not yet released; when it is, swap the source files in `data/boost_factor/profiles/` without further code change.
- **Posterior-mean projection coefficients.** The C19 projection parameters are evaluated at the posterior mean of the upstream chain. Row-by-row covariance propagation is not implemented.
- **Single Y3 realisation.** The §8.1 empirical ratio is computed on one Y3 realisation; Wu+2022 averages 12 Y1 + 1 Y3 realisations and is correspondingly tighter.

10 File map

- `MockDataVector.ipynb` — the production notebook. Top-to-bottom execution reproduces every figure and writes the output archive.
- `src/costanzi_selection.py` — C26 Eq. 15 sampler and C19 Eq. 6 mixture sampler, plus the legacy single-scale $\mathcal{B}_{\rm sel}$ weight.
- `src/stacked_profile_weighted_by_mass_redshift.py` — Wu+22 / Esteves (M, z)-matched random reference builder.
- `src/radial_bins_phys_mpc.py` — the 15-bin physical radial grid `rp_phys_mpc`.
- `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` — 15 posterior samples of the 10 C19 projection coefficients.
- `src/fileLoc.py` — machine-resolved path table for the raw Buzzard halo catalogue, profile output, and $\beta_{\rm eff}$ table.
- `output/MockDataVector.npz` — saved data vector (Tab. 1).
- `output/figs/` — headline plots embedded in this PDF.
- `docs/MockDataVector.tex` — this document.

11 References

- Costanzi et al. 2026, *Forward analytical model for the optical selection bias on galaxy cluster lensing profiles.* Eqs. 15 (mass–richness), Appendix C Eq. C1 (selection-bias fit), Sec. III (Poisson–Gaussian convolution).
- Costanzi et al. 2019, arXiv:1807.07072. Eq. 6 (Gaussian + EMG mixture), Appendix A Eq. A8 (per-parameter forms).
- Wu et al. 2022, arXiv:2203.05416. Fig. 2 (selection-bias panel grid) and Appendix B (mass–redshift weighted reference).
- McClintock et al. 2019, MNRAS 482, 1352 — DES Y1 cluster binning convention used throughout.